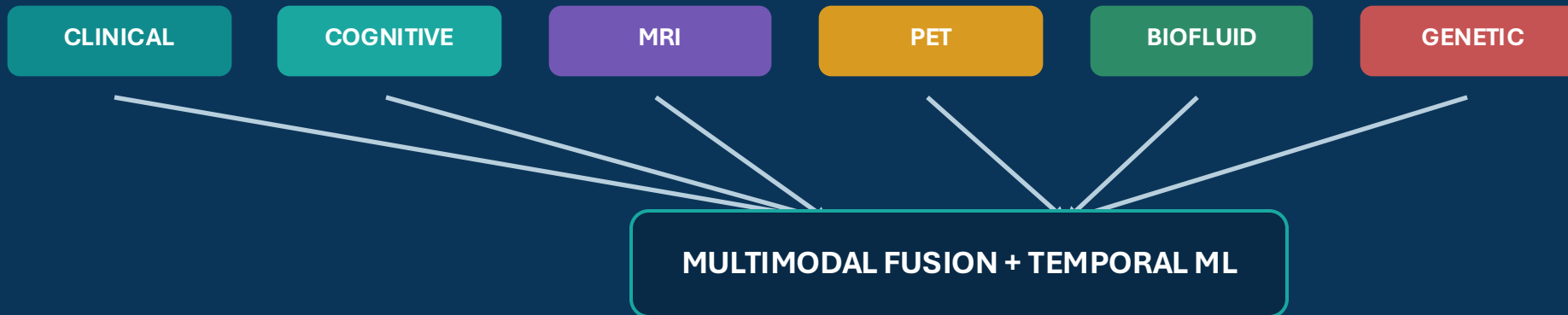


NeuroFusion-AI

Explainable Multimodal Machine Learning for Alzheimer's Disease Classification and Longitudinal Progression Prediction Using ADNI



Project presentation + student implementation roadmap

Research pipeline

A reproducible workflow that constructs ADNI cohorts, harmonizes modalities, prevents temporal leakage and produces auditable datasets.

Predictive models

CN/MCI/AD classification plus prospective MCI→AD progression prediction at defensible follow-up horizons.

Trustworthy AI

Calibration, uncertainty, missing-modality robustness, explainability, subgroup analysis and stress testing.

Software platform

Python modules, experiment manager, provenance tracking and reusable configurations.

Research dashboard

Cohort inspection, trajectories, model comparison, calibration, SHAP summaries and progression forecasts.

What does CN / MCI / AD classification mean?

Task A predicts a participant's current disease-state category from information available at a defined visit.

CN

Cognitively Normal

No significant cognitive impairment according to the diagnostic definitions used for the selected ADNI cohort.

MCI

Mild Cognitive Impairment

Measurable cognitive impairment beyond expected normal aging, without meeting the selected ADNI definition of Alzheimer's dementia.

AD

Alzheimer's Disease

Participant classified in the Alzheimer's disease/dementia category according to the diagnostic rules used in the relevant ADNI phase.

The ML question

Given the participant's multimodal measurements at visit t , which current state is most consistent with the learned patterns: CN, MCI or AD?

Important: this is disease-state classification, not future progression prediction.

The model predicts probabilities — not only a label

For participant feature vector X , each model returns a probability distribution across the three classes.

Example participant features

- Age / education
- MMSE • ADAS-Cog • CDR-SB
- MRI regional measures
- PET summaries
- Biofluid biomarkers
- APOE ϵ 4 information

Multiclass probability model

$$P(\text{CN} | X), P(\text{MCI} | X), P(\text{AD} | X)$$

$$P(\text{CN}|X) + P(\text{MCI}|X) + P(\text{AD}|X) = 1$$

Illustrative output

CN

0.08

MCI

0.76

AD

0.16

Predicted class = $\arg \max \{ P(\text{CN}|X), P(\text{MCI}|X), P(\text{AD}|X) \} = \text{MCI}$

How CN / MCI / AD classification enters the ensemble

All five base learners estimate the same three probabilities. The ensemble combines probability vectors, not hard votes.

LR	Logistic Regression	→ [P(CN), P(MCI), P(AD)]
SVM	RBF Support Vector Machine	→ [P(CN), P(MCI), P(AD)]
RF	Random Forest	→ [P(CN), P(MCI), P(AD)]
XGB	XGBoost	→ [P(CN), P(MCI), P(AD)]
MLP	Neural Network	→ [P(CN), P(MCI), P(AD)]

Three ensemble configurations

A Soft Voting

Mean of the five probability vectors

B Weighted Soft Voting

$$P_e(c|X) = \sum_m w_m P_m(c|X), \quad \sum_m w_m = 1$$

C Stacking

$Z = [P_{LR}, P_{SVM}, P_{RF}, P_{XGB}, P_{MLP}]$

Logistic Regression meta-learner uses out-of-fold predictions.

Core comparison: 5 individual models + 3 ensemble configurations = 8 predictive configurations.

Current-state classification vs future progression

The project separates recognition of the current disease state from prospective prediction of what happens next.

TASK A**CN / MCI / AD classification**

Question: “What is the participant’s current disease state?”

$$X_t \rightarrow \{ \text{CN, MCI, AD} \}$$

Input: measurements available at a defined visit t .
Output: three-class probability distribution.

Evaluation: Macro-F1 • class sensitivity • one-vs-rest AUROC • calibration

TASK B**MCI → AD progression**

Question: “Will a participant with MCI progress to AD within a future horizon?”

$$X_{t \leq t_0} \rightarrow P(\text{MCI} \rightarrow \text{AD within } H \text{ months})$$

Feature window: information at or before index t_0 .
Outcome window: conversion strictly after t_0 .

Candidate H: 12 / 24 / 36 months • selected after follow-up and class-count analysis

Task B is longitudinal and prospective; Task A describes the participant’s state at the current visit.

01

Scientific problem

Alzheimer's disease is heterogeneous and longitudinal. The project asks whether combining complementary evidence improves disease-state modelling and progression forecasting while remaining reliable and interpretable.

Central research question

Can multimodal machine learning that integrates heterogeneous and longitudinal ADNI information improve CN/MCI/AD classification and MCI→AD progression prediction while remaining calibrated, interpretable and robust to missing modalities?

H1 • Fusion

Multimodal models should be compared against the best single-modality baseline.

H2 • Time

Longitudinal representations should be tested against baseline-only representations.

H3 • Architecture

Early, late and learned hybrid fusion should be compared experimentally.

H4 • Robustness

Missing-modality handling and uncertainty should be explicitly evaluated.

Two primary prediction tasks

Task A — Disease-state classification

Input: information available for a participant at a defined visit.

Output: probability distribution over CN, MCI and AD.

Baselines: multinomial logistic regression, SVM, Random Forest, XGBoost/LightGBM, MLP.

Key metrics: Macro-F1, class sensitivity, one-vs-rest AUROC, calibration.

Task B — MCI→AD progression

Cohort: participants with MCI at index time t_0 .

Output: probability of future conversion after t_0 .

Candidate horizons: 12, 24 and 36 months, selected after coverage analysis.

Key metrics: AUROC, AUPRC, sensitivity, specificity, Brier score and calibration.

CN / MCI / AD classification — what exactly is predicted?

Question being answered

“Given the information available for this participant at this visit, which current diagnostic state is most consistent with the data?”

Three classes

CN = Cognitively Normal

MCI = Mild Cognitive Impairment

AD = Alzheimer’s Disease

Input X may combine clinical, cognitive, MRI, PET, biofluid and genetic features from the same permitted time point.

Model output: a probability distribution

$P(\text{CN} | X)$, $P(\text{MCI} | X)$, $P(\text{AD} | X)$

$$P(\text{CN}|X) + P(\text{MCI}|X) + P(\text{AD}|X) = 1$$

Example

CN = 0.08 MCI = 0.76 AD = 0.16

Predicted class = $\arg \max \{0.08, 0.76, 0.16\} = \text{MCI}$

Why keep probabilities? They enable calibration, uncertainty analysis and ensemble combination; a hard label alone loses this information.

MCI → AD progression — predicting what happens next

Who enters this experiment?

Only participants who have MCI at the index time t_0 .

Prediction question

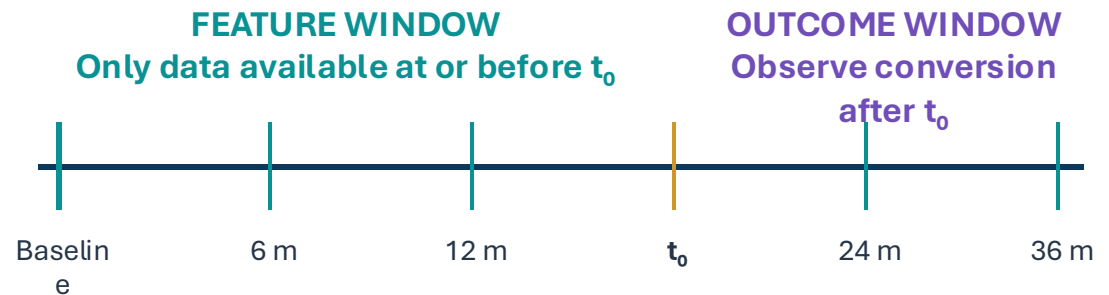
“What is the probability that this participant will convert from MCI to AD within a future horizon H ?”

Formal form

$P(\text{MCI} \rightarrow \text{AD within } H \text{ months} \mid X \leq t_0)$

Candidate H : 12, 24 or 36 months — chosen only after checking follow-up coverage and event counts.

Leakage-safe temporal logic



Critical rule: the model must never use measurements collected after t_0 to predict an outcome after t_0 .

Example: if the 36-month model outputs 0.72, the estimated probability of MCI→AD conversion within 36 months is 72% — subject to calibration and uncertainty.

Current-state recognition vs future-risk forecasting

TASK A — CN / MCI / AD

Type: 3-class classification

Population: eligible ADNI participants

Time: defined visit

Output: P(CN), P(MCI), P(AD)

Purpose: model the participant's current disease state

TASK B — MCI → AD

Type: future-event prediction

Population: MCI participants at t_0

Time: horizon H after t_0

Output: P(conversion within H)

Purpose: forecast longitudinal progression

Key evaluation acronyms

Macro-F1 — average F1 across CN/MCI/AD, giving each class equal importance.

AUROC — Area Under the Receiver Operating Characteristic curve; measures discrimination across thresholds.

AUPRC — Area Under the Precision–Recall Curve; especially informative when progression events are imbalanced.

Brier score — mean squared error of predicted probabilities; lower is better. Calibration asks whether predicted risks match observed frequencies.

02

ADNI data foundation

We must first understand the dataset, access rules, identifiers, phases, visits, modality coverage and data dictionaries before training any model.

Build a participant × visit longitudinal table

Each row represents one participant at one visit. The fields below identify the person, the visit/time point, the diagnostic state, cognitive scores and available biomarker/imaging information.

EXAMPLE RECORD

RID	VISCO DE	EXAMDAT E	DX	MMS E	ADAS1 3	CDRS B	MRI	PET	CSF	APOE
001	m24	2026-...	MCI	24	19	3.0	✓	✓	✓	0/1/2

The ✓ symbols indicate that the corresponding modality/measurement is available at that visit.

IDENTIFIERS & DIAGNOSIS

RID — Research/participant identifier used to link records.
VISCODE — Visit code, e.g. bl = baseline, m12 = month 12.
EXAMDATE — Date of the examination/assessment.
DX — Diagnosis: CN, MCI or AD.

COGNITIVE / CLINICAL MEASURES

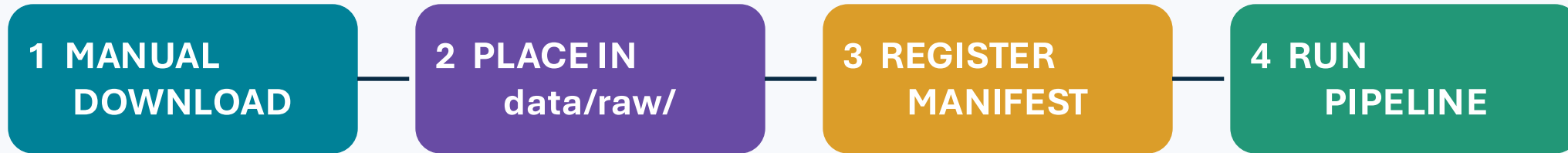
MMSE — Mini-Mental State Examination: brief global cognitive screening score.
ADAS13 — 13-item Alzheimer’s Disease Assessment Scale–Cognitive Subscale: measures cognitive impairment; higher scores generally indicate greater impairment.
CDRSB — Clinical Dementia Rating–Sum of Boxes: summarizes impairment across cognitive and functional domains; higher values indicate greater severity.

IMAGING, BIOMARKERS & GENETICS

MRI — Magnetic Resonance Imaging: structural brain imaging; the core project uses processed numerical brain-region measures.
PET — Positron Emission Tomography: molecular/functional imaging, including FDG, amyloid or tau PET where available.
CSF — Cerebrospinal Fluid: fluid biomarkers related to Alzheimer’s pathology.
APOE — Apolipoprotein E genotype; ε4 is an established genetic risk factor. Encode the exact variable definition used by the selected ADNI release.

ADNI data are provided manually — not downloaded by the software

Students obtain the approved files outside NeuroFusion-AI. The implementation begins only after the files are available locally.



No IDA API • No web scraping • No automated authentication • No automated downloading

The student records the file name, ADNI phase/release, download date, processing version and dictionary used. From this point onward, the pipeline must be reproducible.

IMPLEMENTATION BOUNDARY: local files → harmonization → cohort → modelling

Minimum ADNI data package for the core project

Use processed/tabular information first. Raw 3D MRI/PET and high-dimensional omics are optional extensions.

Diagnosis + demographics

Clinical + cognitive

APOE

Biofluid biomarkers

MRI numerical summaries

PET numerical summaries

Required identifiers/metadata: RID or PTID • VISCODE • EXAMDATE • diagnosis • modality availability • processing/version information

Core thesis first → optional raw imaging/omics only if time and cohort size justify it

Organize the data before modelling

Recommended project structure

```
NeuroFusion-AI/  
  data/  
    raw/  
      demographics/  
      diagnosis/  
      cognitive/  
      clinical/  
      genetics/  
      biofluid/  
      mri_numeric/  
      pet_numeric/  
    dictionaries/  
    interim/  
    processed/  
  configs/  
  src/  
  experiments/  
  models/  
  reports/  
  notebooks/
```

Rule 1

Never manually edit files in data/raw/. Treat raw downloads as immutable evidence.

Rule 2

Every processed dataset must be reproducible from code + configuration.

Rule 3

Track table/file version, download date, variable dictionary and processing provenance.

Rule 4

Keep identifiers only in the approved controlled research environment.

Build a participant × visit longitudinal table

RID	VISCODE	EXAMDATE	DX	MMSE	ADAS13	CDRSB	MRI...	PET...	CSF...	APOE
001	bl	...	CN	29	8	0	✓	–	–	0
001	m12	...	MCI	26	14	1.5	✓	✓	–	0
001	m24	...	MCI	24	19	3	✓	✓	✓	0
001	m36	...	AD	20	28	6	✓	✓	✓	0

Join carefully using participant and visit/time information. RID alone is not enough for longitudinal alignment.

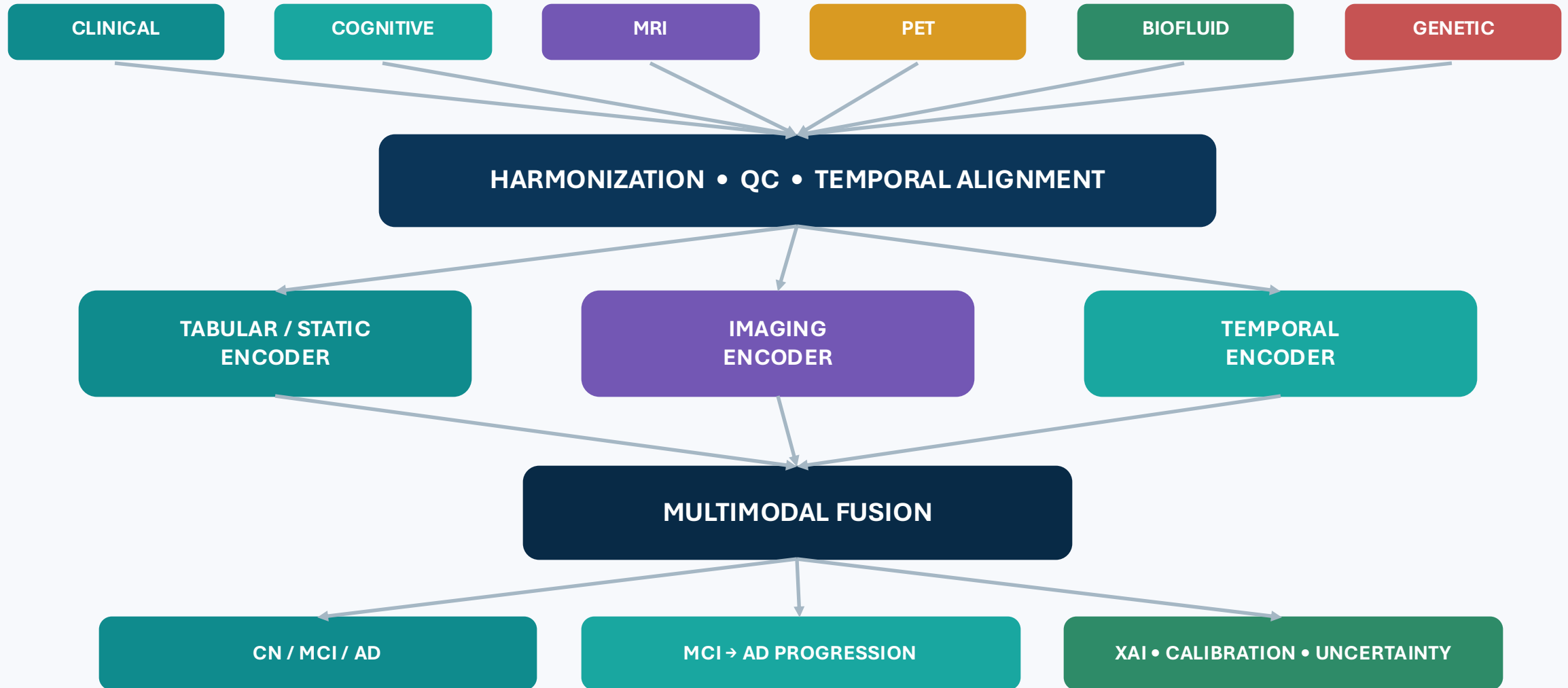
The exact variables and joins must be documented from the approved ADNI release used by the student.

03

System architecture

Separate data governance, harmonization, representation, fusion, prediction and trustworthiness. This makes leakage prevention and reproducibility first-class components.

NeuroFusion-AI end-to-end architecture



Python software modules

Data Import & Harmonization

Read local files; validate schema; harmonize variables; provenance

Patient Timeline Engine

Visit and timestamp alignment

MRI/PET Engines

Numeric summaries; optional 3D encoders

Missing Data Engine

Imputation, masks, completeness

Base Learner Engine

LR, SVM, RF, XGBoost, MLP

Calibration / Uncertainty

Probability reliability and uncertainty

Experiment Manager

Splits, seeds, configs, metrics

Cohort Builder

Eligibility, labels, index dates, horizons

Clinical/Cognitive Engines

Feature construction and trajectories

Biomarker/Genetic Engines

Biological feature processing

Multimodal Fusion Engine

Early, late, hybrid fusion

Ensemble & Progression Engine

Voting, weighted voting, stacking; MCI→AD prediction

Explainable AI Engine

Global, local and modality explanations

Dashboard

Research visualization and inspection

Recommended technology stack — with legend

The technologies below are recommended implementation tools. The legend explains what each one is and its role in NeuroFusion-AI.

AREA	TECHNOLOGY	WHAT IT IS	ROLE IN THIS PROJECT
Data	Python	General-purpose programming language.	Main implementation language for the complete pipeline.
	Pandas / NumPy / SciPy	Libraries for tables, numerical arrays and scientific computing.	Data cleaning, transformations, statistics and feature preparation.
Machine Learning	scikit-learn	Python ML library for preprocessing, models and evaluation.	LR, SVM, RF, pipelines, cross-validation and metrics.
	XGBoost / LightGBM	Gradient-boosted decision-tree frameworks.	Strong nonlinear tabular baseline models.
Deep Learning	PyTorch	Deep-learning framework using tensors and automatic differentiation.	MLP, temporal models and learned multimodal fusion.
Neuroimaging	MONAI / NiBabel	Medical-imaging DL toolkit / neuroimaging file I/O library.	Optional advanced processing of 3D MRI/PET images.
Explainability	SHAP / Integrated Gradients	Methods that attribute model predictions to input features.	Global/local explanations for tree and neural models.
Storage	Parquet / PostgreSQL	Columnar file format / relational database system.	Processed analytical datasets, metadata and provenance.
Services	FastAPI	Python framework for building web APIs.	Optional service layer exposing model/data functions.
Dashboard	Plotly / Dash	Interactive plotting library / Python dashboard framework.	Research dashboard, trajectories, comparisons and forecasts.
Experiments	MLflow	Experiment tracking and model-management platform.	Optional registry for parameters, metrics, runs and artifacts.

ACRONYM / TERM LEGEND ML = Machine Learning • LR = Logistic Regression • SVM = Support Vector Machine • RF = Random Forest • MLP = Multilayer Perceptron • API = Application Programming Interface • MRI = Magnetic Resonance Imaging • PET = Positron Emission Tomography

04

Implementation roadmap

Implement in layers: local data → cohort → five base learners → ensemble comparison → longitudinal prediction → trustworthiness → dashboard → reproducibility package.

Phase 0 — Project setup and governance

- Confirm the approved ADNI files have been downloaded manually and stored securely.
- Create Git repository, Python environment and project directory structure.
- Place the data dictionary and selected ADNI files in the agreed local data folders.
- Create a local provenance manifest: file, phase/release, download date, processing version, checksum if practical.
- Define coding conventions, configuration format, random-seed policy and experiment naming.

Exit criterion

A fresh clone can create the environment; local ADNI files are discoverable through configuration; raw data are not committed to public version control.

Phase 1 — Cohort construction

- Import the provided diagnosis, demographics and core cognitive tables first.
- Normalize participant identifiers, visit codes and dates without destroying original values.
- Create a canonical visit timeline for every participant.
- Define CN/MCI/AD label rules and document phase-specific diagnosis mappings.
- Produce cohort attrition tables: downloaded → eligible → included → excluded, with reasons.
- Add modality-availability flags for every participant/visit.

Deliverable D1

Data & cohort specification containing variables, inclusion/exclusion rules, labels, horizons and provenance.

Phase 2 — Five base learners

Train the same five model families under the same patient-level split and preprocessing protocol.

**Logistic
Regression**

M1

linear + transparent

SVM (RBF)

M2

nonlinear margin

**Random
Forest**

M3

bagging / trees

XGBoost

M4

gradient boosting

MLP

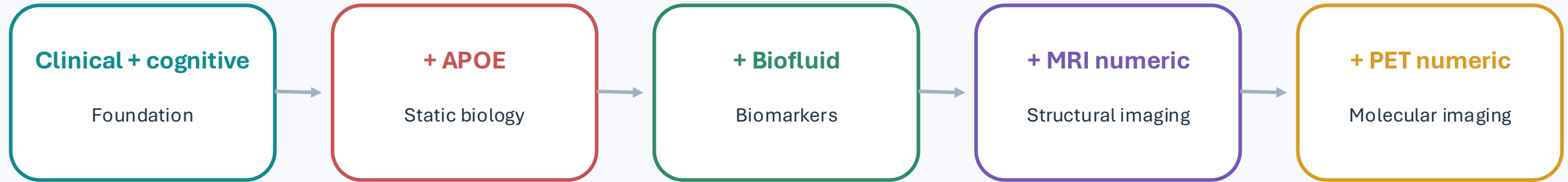
M5

neural nonlinear

Every model outputs probabilities — not only hard class labels

Fit imputation, scaling and feature selection inside training folds only. Hyperparameters are tuned without using the final test set.

Phase 3 — Build the multimodal feature space



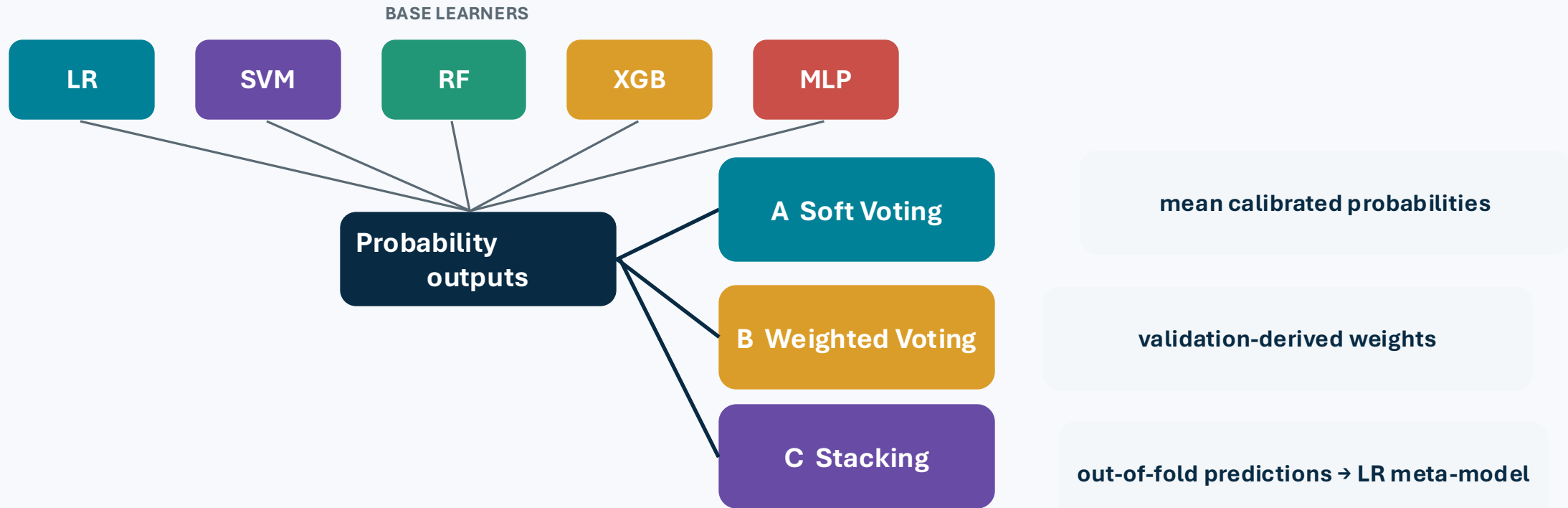
At each step, record sample size, missingness, class balance and performance. A modality is not “useful” merely because it exists.

Decision rule

Expand only when the remaining cohort is large enough to support a defensible experiment.

Phase 4 — Ensemble methodology

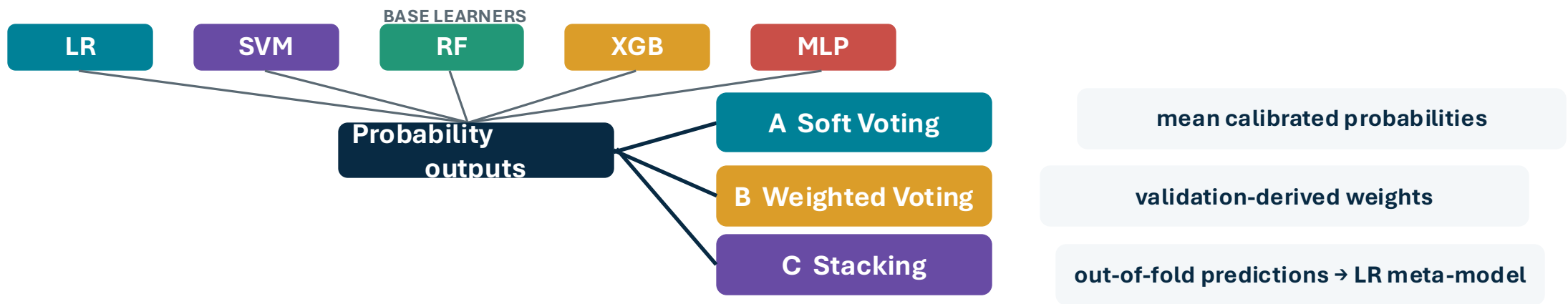
Five base learners generate probability vectors. Compare three ways of combining them.



Compare: 5 individual models + 3 ensemble configurations = 8 primary predictive configurations.

Phase 4 — Ensemble methodology: diagram explained

The original diagram is retained; the legend below explains what every box and connection represents.



Compare: 5 individual models + 3 ensemble configurations = 8 primary predictive configurations.

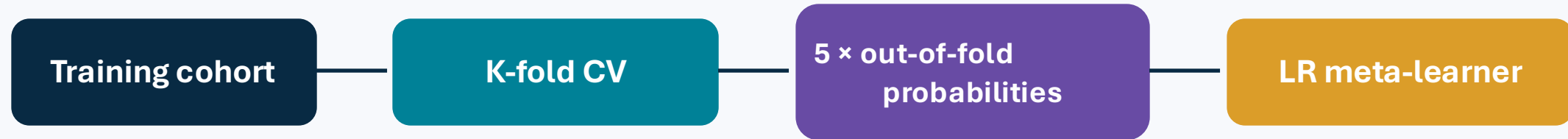
HOW TO READ EACH BOX IN THE DIAGRAM

LR — Logistic Regression	Linear probabilistic baseline; estimates class or progression probabilities.	A — Soft Voting	Mean calibrated probabilities: all five base learners contribute equally.
SVM — Support Vector Machine	RBF-kernel nonlinear classifier; probability estimates are calibrated before combination.	B — Weighted Voting	Validation-derived weights: models can contribute differently; weights are chosen without the test set.
RF — Random Forest	Bagging ensemble of decision trees; its tree votes are converted into probabilities.	C — Stacking	Out-of-fold base-model probabilities become inputs to a second-level Logistic Regression meta-model.
XGB — XGBoost	Gradient-boosted decision trees trained sequentially to reduce prediction errors.	Out-of-fold (OOF)	A training participant is predicted by a fold model that did not train on that participant.
MLP — Multilayer Perceptron	Feed-forward neural network that learns nonlinear feature relationships.	LR meta-model	A Logistic Regression model learns how to combine the five base-model probability outputs.
Probability outputs	Common interface: every base learner produces probabilities rather than only a hard class label.	8 configurations	The study compares 5 individual models plus 3 ensemble strategies under the same evaluation protocol.

RBF = Radial Basis Function • Bagging = bootstrap aggregation • OOF = out-of-fold • Meta-model = model that combines other models' outputs

Leakage-safe stacking

Stacking is the advanced ensemble: the meta-learner sees only out-of-fold predictions from the five base models.



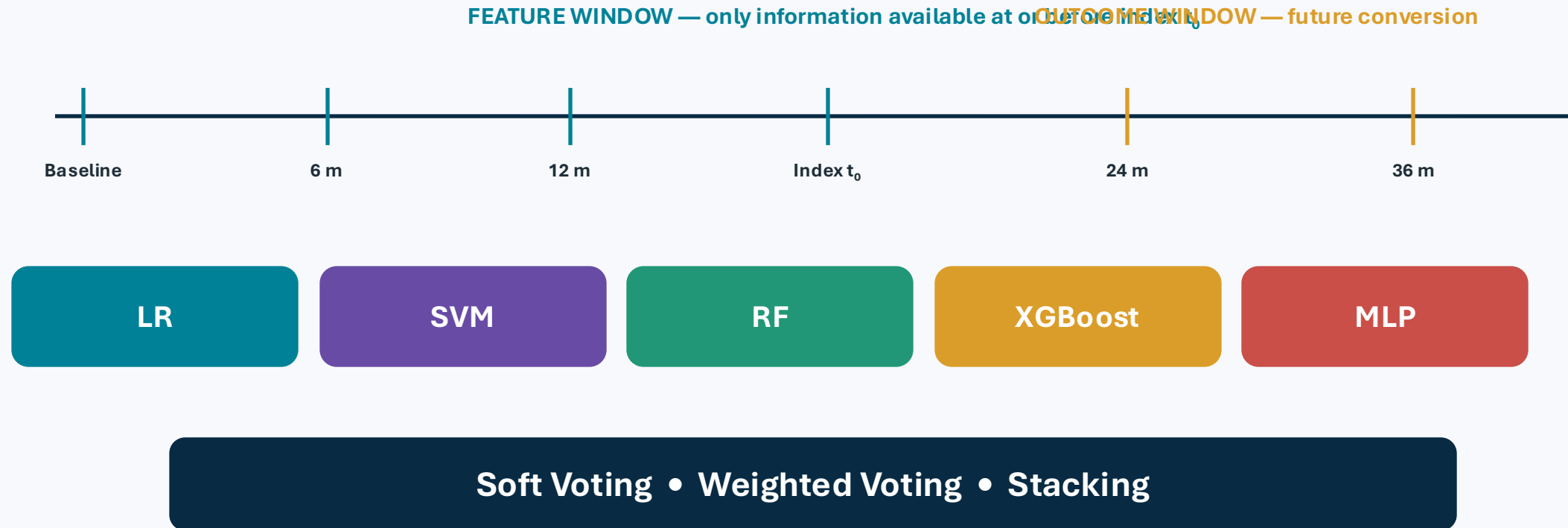
$$\mathbf{Z} = [P_{LR}, P_{SVM}, P_{RF}, P_{XGB}, P_{MLP}]$$

After the meta-model is fixed, refit the five base learners on the permitted training data and evaluate the complete ensemble once on the untouched test participants.

TEST SET IS NEVER USED TO CHOOSE BASE MODELS, WEIGHTS OR META-MODEL

Phase 5 — Ensemble MCI→AD progression

Reuse the five base learners and three ensemble strategies for future conversion risk.



Primary horizon: select after cohort inspection; 36 months is a candidate, with 12/24 months added only if follow-up and class counts are adequate.

Phase 6 — Calibration, uncertainty and XAI

Calibration

Evaluate whether predicted probabilities correspond to observed frequencies.
Candidate methods: Platt scaling, isotonic regression or temperature scaling, fitted on validation data only.

Uncertainty

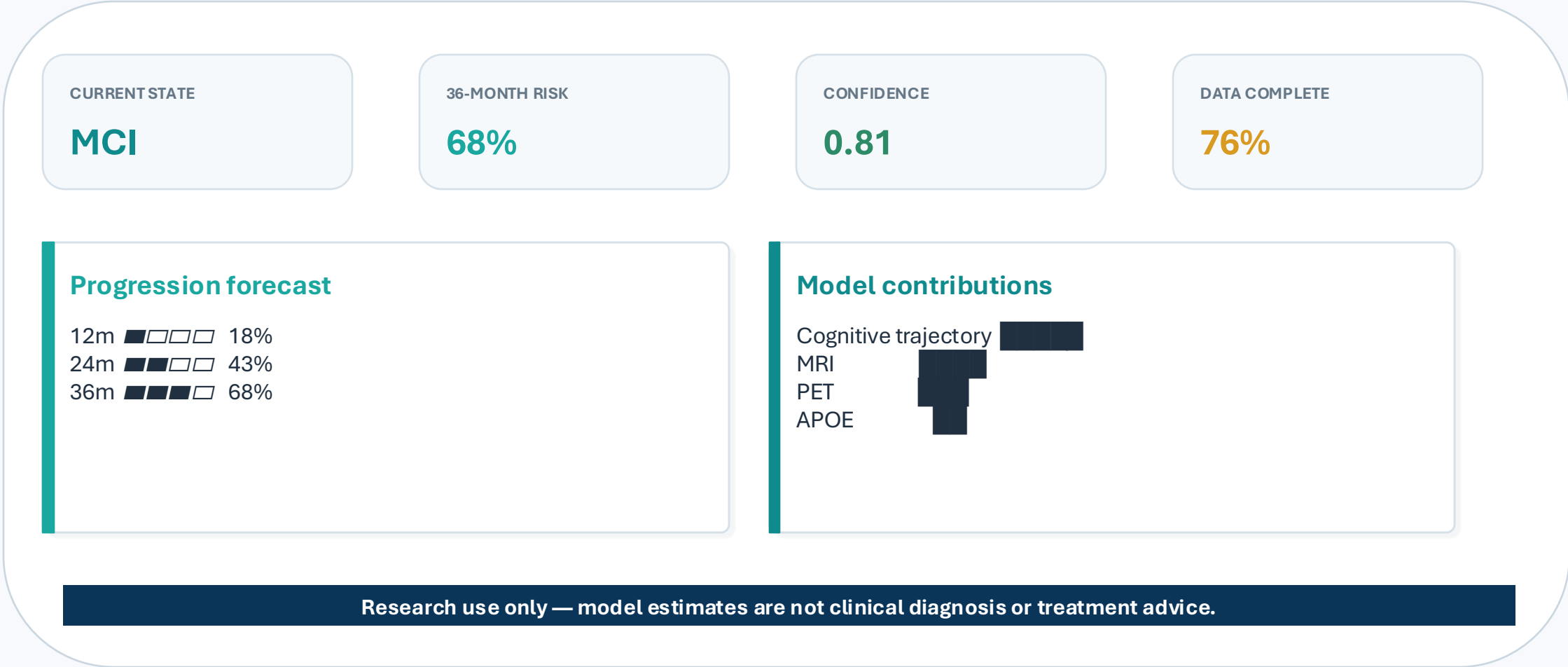
Attach probability reliability, data-quality and modality-availability information to predictions. Define confidence explicitly; do not invent a cosmetic score.

Explainability

Use SHAP for tree-based models; integrated gradients or ablation for neural models. Describe model contributions—not biological causation.

Trustworthiness is part of the experimental programme, not a dashboard decoration.

Phase 8 — Research dashboard



05

Experimental discipline

A defensible thesis depends on leakage-safe splitting, nested preprocessing, consistent held-out participants, transparent attrition and reproducible experiment metadata.

Leakage prevention checklist

- ✓ Split at participant level — repeated visits/scans never cross partitions.
- ✓ Fit imputation, scaling and feature selection inside training folds only.
- ✓ Fit calibration using validation data, never the final test set.
- ✓ For progression, enforce timestamp cut-offs programmatically.
- ✓ Use the same held-out participants for model comparisons whenever possible.
- ✓ Persist seeds, cohort query/version, preprocessing parameters and model configuration.
- ✓ Maintain a prediction ledger with model version, modality mask and probabilities.

Experimental programme

The thesis should answer a small number of explicit questions rather than accumulate algorithms.

E1

Single-modality benchmarks

E2Multimodal five-model
benchmark**E3**

Soft voting

E4

Weighted soft voting

E5

Stacking

E6MCI→AD ensemble
progression**E7**

Missing-modality stress test

E8

Calibration + uncertainty

E9

Explainability + ablation

E10

Subgroup analysis

Main comparison: best individual learner vs three ensembles on identical held-out participants

Evaluation framework

TASK	PRIMARY METRICS	SECONDARY ANALYSES
CN/MCI/AD	Macro-F1; class sensitivity; one-vs-rest AUROC	Confusion matrix; calibration
MCI→AD	AUROC; AUPRC; sensitivity; specificity	Brier score; calibration; PPV/NPV
Fusion	Δ performance vs best single modality	Paired bootstrap confidence intervals
Robustness	Performance under missing modalities	Degradation curves
Subgroups	Metrics by prespecified strata	Uncertainty + sample-size reporting

Accuracy alone is insufficient. Principal results should include confidence intervals.

Deliverables and thesis outputs

D1 — Data & cohort specification

variables, rules, labels, horizons, provenance

D3 — Baseline benchmark

classical ML + single-modality results

D5 — Longitudinal predictor

MCI→AD models at defensible horizons

D7 — Research dashboard

interactive experiment + participant views

D2 — Reproducible Python pipeline

import, harmonization, QC, features, splits

D4 — Multimodal models

early, late and hybrid fusion

D6 — Trustworthiness suite

calibration, uncertainty, XAI, subgroup, robustness

D8 — Master's thesis + reproducibility package

methods, experiments, results, limitations, reproducibility

Success criteria

- A leakage-safe, documented cohort can be reconstructed from the manually provided ADNI files using code and configuration.
- All five base learners and all three ensemble strategies are evaluated on the same leakage-safe protocol.
- MCI→AD prediction is evaluated at one or more defensible horizons.
- Calibration and uncertainty are assessed—not only classification accuracy.
- Missing-modality and modality-ablation experiments are completed.
- Participant-level explanations and model provenance are available in the dashboard.
- The student can explain limitations, selection bias, scanner/site heterogeneity and why the system is research support—not a clinical diagnostic device.

First actions for students

Assume the approved ADNI files have already been downloaded manually.

1

Place files

Organize approved ADNI tables under data/raw/; keep them immutable.

2

Create manifest

Record file/release/date/version/dictionary and modality.

3

Harmonize

Normalize identifiers, visits, dates and variable names in code.

4

Build cohort

Create participant × visit table and CN/MCI/AD labels.

5

Split safely

Create participant-level train/validation/test partitions.

6

Train 5 models

LR, SVM, Random Forest, XGBoost and MLP.

7

Build ensembles

Soft voting, weighted voting and stacking.

8

Evaluate

Classification → progression → calibration → XAI → robustness.

LOCAL DATA → COHORT → 5 BASE LEARNERS → 3 ENSEMBLES → PROGRESSION → TRUSTWORTHINESS

Key sources and documentation

Project basis

- NeuroFusion-AI Master's Project Brief, v2 — uploaded project specification.

Official ADNI resources

- ADNI Data and Samples / Data Access
- ADNI Data User Guide / Study Files
- ADNI Data Dictionary documentation
- ADNI Clinical Assessments
- ADNI Genetics & Related Omics
- LONI Image & Data Archive (IDA) user documentation

Important implementation note

Exact ADNI files, variables, cohort counts and biomarker definitions must be documented from the approved data release actually used by the student.

Do not hard-code assumptions from older papers or tutorials without checking the current dictionary and release.